

Algorithm Evaluation Criteria

The purpose and use of fraud detection algorithms and models is to detect and prevent fraud in banking transactions, which necessitates the adoption and use of the most effective model. Different elements form the basis for the criteria that professionals and institutions use to assess the effectiveness of fraud detection models. One of the foremost elements of the criteria to assess the effectiveness of fraud detection models is accuracy, which is however, limited in applicability because it may lead to misleading conclusions in highly imbalanced datasets (Hashemi, Mirtaheri and Greco, 2022). Precision is another important parameter, which measures the proportion of detected fraud cases that are genuinely fraudulent, with higher precision rates resulting in reduced cases of unnecessary fraud investigations and customer inconvenience (Al-dahasi et al., 2025). Another critical element is recall, which is important because it measures the proportion of actual fraud cases successfully identified, making it very important because missed fraudulent transactions have substantial financial consequences (Islam et al., 2024). Another consideration in assessing the effectiveness of fraud detection models is the F1-score, which balances precision and recall, making it particularly appropriate for highly imbalanced fraud datasets, making it a more reliable metric than accuracy alone (Chen, 2025). The ROC-AUC, another parameter, evaluates a model's overall discriminatory ability across multiple classification thresholds, with ensemble algorithms consistently achieving higher ROC-AUC values than traditional classifiers (Hashemi, Mirtaheri and Greco, 2022).

Another element of the evaluation criteria is the false positive rate because excessive false positives increase investigation costs, inconvenience customers and reduce trust in fraud detection systems, which means that models should be capable of maintaining high recall while minimising false positives (Li and Chen, 2025). Detection Latency as a metric is a critical

consideration in real-time fraud detection because fraud detection requires rapid model prediction with minimal transaction delays. Li and Chen (2025) observed that computationally intensive models might achieve higher predictive performance but exhibit unacceptable processing latency for operational banking environments. Computational Efficiency is another critical evaluation element because efficient algorithms reduce processing time and hardware requirements while supporting continuous fraud monitoring, with LightGBM and XGBoost models demonstrating an ability to balance predictive performance with computational efficiency (Hashemi, Mirtaheri and Greco, 2022). Scalability entails ensuring that fraud detection models maintain desired levels of performance when processing millions of transactions daily, with ensemble learning and distributed machine learning architectures demonstrate improved scalability for large banking systems (AbouGrad and Sankuru, 2025). Finally, Explainability and Interpretability, is a critical parameter because financial institutions increasingly require interpretable models to satisfy regulatory requirements and facilitate fraud investigations. Although deep learning and ensemble models provide superior predictive performance, their limited explainability remains a major challenge for practical deployment (Alrasheedi, 2025). Independently, and collectively, the parameters help users to assess the effectiveness of the models to predict, and therefore, forestall fraud in financial transactions.

Datasets Used in Fraud Detection Studies

Fraud detection research primarily relies on publicly available datasets due to the confidentiality of real banking transaction data (Prabowo, 2016; Hernandez Aros et al., 2024). Further, a review of literature has demonstrated that it is important to review and evaluate multiple datasets and algorithms to improve the robustness and generalisability of fraud detection models (AbouGrad and Sankuru, 2025; (Dichev, Zarkova and Angelov, 2025; Du et al., 2024). Larger datasets improve model generalisation but increase computational

requirements (Du et al., 2024). Public fraud datasets vary from several hundred thousand to millions of transaction records (AbouGrad and Sankuru, 2025; Qi, 2025). From the extant literature, it is evident that the most widely used datasets include the Credit Card Fraud Dataset, IEEE-CIS Fraud Detection Dataset, BankSim Dataset, PaySim Dataset, and the Bank Account Fraud Dataset Suite (NeurIPS 2022) (Almazroi & Ayub, 2023; AbouGrad & Sankuru, 2025). Evidently, there are multiple publicly available datasets for use in fraud-detection algorithms' testing.

Datasets Evaluation and Selection

Multiple datasets exist, from which users can select and use data for ML training. Users must critical to evaluate the datasets before settling on a final one for subsequent application. Various factors influence the appropriateness of datasets. One of the key factors is the Fraud-to-Legitimate Transaction Ratio, with multiple fraud datasets exhibiting severe class imbalance, with fraudulent transactions typically representing less than 1% of observations (Hashemi, Mirtaheri and Greco, 2022). This imbalance significantly influences model evaluation and algorithm selection. Another factor is the feature types in each dataset. Current fraud datasets include transactional variables (amount, time, merchant), behavioural variables (transaction frequency, spending behaviour), and demographic characteristics (Qi, 2025). Notably though, the sets with richer features generally improve predictive performance. Data quality and completeness is another critical consideration; missing values, inconsistent records and noisy transaction data reduce model effectiveness. In such datasets, extensive pre-processing and feature engineering remain essential components of fraud detection research (Du et al., 2024). Additionally, the availability of labelled fraud cases is a critical consideration in evaluating datasets for their appropriateness in use in ML for fraud detection training. Supervised learning requires accurately labelled fraudulent transactions. As AbouGrad and Sankuru (2025)

observed, limited availability of labelled fraud cases motivates increasing use of unsupervised and semi-supervised learning approaches.

From a consideration of these factors, it is possible to assess datasets for suitability for supervised and unsupervised learning. The Credit Card Fraud Dataset and IEEE-CIS Dataset are primarily suited for supervised learning because fraud labels are available. The Bank Account Fraud Dataset Suite also supports anomaly detection and unsupervised experimentation owing to its behavioural characteristics (AbouGrad and Sankuru, 2025). The publicly available datasets facilitate benchmarking, reproducibility and algorithm comparison across studies. However, many datasets represent specific institutions, countries or fraud types, limiting the generalisability of findings to broader online banking environments (Dichev, Zarkova and Angelov, 2025). Therefore, it is important to validate fraud detection models using multiple datasets to foster robustness and external validity.

Identification and Evaluation Publicly Available Fraud Detection Datasets

Public datasets provide an accessible platform for benchmarking fraud detection algorithms while ensuring reproducibility of research findings (Prabowo, 2016). The Credit Card Fraud Dataset remains the most frequently used benchmark because of its labelled transactions and severe class imbalance, making it suitable for evaluating fraud detection techniques (Hashemi et al., 2022; Zhang et al., 2022). The IEEE-CIS Fraud Detection Dataset offers richer customer and transaction features but requires extensive preprocessing due to missing values and high dimensionality (Almazroi & Ayub, 2023). The Bank Account Fraud Dataset Suite (NeurIPS 2022) better reflects modern online banking fraud by incorporating account-level behavioural characteristics and realistic fraud scenarios (AbouGrad & Sankuru, 2025). Synthetic datasets such as BankSim enable large-scale experimentation but may not fully capture the complexity of real-world fraud behaviour (Zhang et al., 2023). Therefore, it is evident that no single dataset

represents all forms of online banking fraud sufficiently, which brings to the fore the importance of using multi-dataset validation.

Selection of the Most Appropriate Dataset for the Study

Study objectives, fraud type, feature availability, and representativeness of online banking environments should guide dataset selection (Compagnino et al., 2025). The Bank Account Fraud Dataset Suite is well suited for evaluating account-based online banking fraud because it reflects contemporary banking transaction characteristics (AbouGrad & Sankuru, 2025). The Credit Card Fraud Dataset remains valuable for benchmarking algorithm performance owing to its extensive use in prior studies and availability of labelled fraud cases (Hashemi et al., 2022). Combining the two datasets allows evaluation of algorithm robustness across different fraud scenarios and transaction characteristics. Few studies evaluate models across multiple datasets, creating uncertainty regarding model generalisability across financial institutions (Dichev et al., 2025). As such, future studies should incorporate multiple datasets to improve external validity and model transferability.